Milestone 1: Define, Understand Users, and Set Usability Goals

Purpose: In this milestone, you will define your interactive app, identify your target users (via a proto-persona), set measurable usability goals, learn from a similar interface using usability heuristics, and plan for future usability testing and an A/B test.

Deliverables (submit as one PDF):

1. Project Concept
2. Proto-Persona
3. Usability Targets (ISO + Five Components)
4. Core Design Principles
5. Heuristic Evaluation of a Similar Interface
6. Usability Test Plan (3 tasks + 1 A/B test plan)

1. Product Overview (20 pts)

Begin your submission with an overview section that includes two parts: a brief snapshot followed by one paragraph. These two parts work together—the snapshot quickly defines your product at a glance, while the paragraph explains the idea in context and demonstrates your understanding of the difference between UI and UX.

Part A: Project Concept Statement (10 pts)

Include all four items:

- **Product Name:** A working name for your app (it may change later).
  - AirAware
- **Target Users:** The specific group your app is designed for
  - Travelers who experience significant travel stress & travel often for leisure to see their family & friends
- **Core Use:** In one sentence, the single main job your app helps users accomplish.
  - Our app helps users who travel regularly quickly understand the likelihood of their flight getting delayed, and the reasoning behind it in clear language, so that they can plan their visits to see their family & friends in a stress-free manner.
- **Problem Solved:** In one sentence, the real-world pain point your target users face today that your app addresses.
  - The real-world pain point our target users face today that our app addresses is that there is an app called Flighty that exists that offers high-speed flight tracking, but Flighty is expensive, with subscription models that are too expensive for users who travel more casually such as to visit family & friends. These users are missing out on the ability to track the probability of flight delays, so they experience them at a

higher rate than more frequent premium flyers who own Flighty, despite the fact that they have more flexibility around their departure and arrival dates, since they typically are not flying for business.

- **Design Vision:** In 1–2 sentences, what the experience should feel like for the user (for example: clear, fast, guided, confidence-building, low-friction).

  - The experience should feel stress-free since these users are traveling for enjoyment. It should quickly provide users with the understanding of the likelihood of their delay, the reason behind it, and notify users if their flight is of high risk of delay along with the soonest alternative flights with a lower likelihood of delay.

Part B: Concept Narrative (10 pts)

Write one concise paragraph that explains what your interactive app is, what problem it solves, and who it is for. In the same paragraph, you must demonstrate the distinction between UI and UX. Your UI description should briefly describe the likely screens/pages and key interface components at a high level (for example: a dashboard, a browse/search view, a details view, and a create/edit form; navigation, filters, forms, confirmations). This is an overview only and may change later. Your UX description should describe what using the app should feel like and how the user moves through the core flow with minimal confusion, minimal effort, and strong confidence. Focus on the journey from goal to completion, how friction is reduced, and how mistakes are prevented or recoverable.

Our interactive app is an affordable, stress-free app that helps travelers quickly understand the likelihood that their flight gets delayed, and the reason behind it in clear language, so that they can visit their friends and family without unnecessary airport travel stress. Our app solves the problem that the Flighty iOS app follows expensive subscription models, leaving more casual travelers without access to accurate high-speed simply-written information regarding the likelihood of their flight being delayed. It also helps users find an alternative flight quickly that minimizes the likelihood of a flight delay. Our UI will likely include a risk-level in the top-left corner for their current flight such that it is the first thing that they see as the likelihood of delay is the key priority for our users to see. This will likely include a percentage & short description of historic delay times, weather risks, and any reasons behind current risk levels in clear language. Additionally, it will be colored red, yellow, or green depending on risk-level to grab the user's attention. It will also include alert banners for any significant change in flight delay likelihood, along with an alternative flight with low likelihood of delay to help minimize stress. The application will feature light blue features & calm clouds, reflecting calm air flights, and images of happy travel with friends & family. The UX should allow for seamless ability to search for a given flight & its risk-level, the ability to filter & sort by risk-level, origin & destination, arrival and departure times, and flight prices. The UX should also allow the user to intuitively see weather conditions on a map interactively with the ability to alter day and time, and general weather forecast information hourly, weekly & daily for a given location in a

separate tab. There will also be easily accessible help documentation & a chatbot to help answer any questions the user may have regarding using the app, though, the app will be intuitive to begin with, following Norman's design principle of affordance, & Nielsen's principle of learnability. Lastly, the dashboard will include error prevention, for example, by disabling the submit button until required fields are complete.

2. Proto-Persona (20 pts)

A proto-persona is an assumption-based user profile. Unlike research-based personas, proto-personas are created using your best understanding of your target users. This is appropriate when direct user research isn't feasible.

Your proto-persona must include:

- **Persona name + photo placeholder:** Name, Age, Role/context (student, parent, employee, etc.), tech comfort level (Low / Medium / High)
    - Isaac Markman, 22, MSDS Student at the University of Virginia, high tech comfort
    -
- **Background:** Who they are and what their day looks like related to the product
    - Isaac is a first-year master's student who frequently travels and hosts his family, friends, and his girlfriend, all who are in different states. Because he is busy with school, his workload, and his own time, every trip is more important. He typically plans his trips ahead of time, checking flight statuses and additional factors (like weather), and often is flying through busy airports. When delays or cancellations occur, they are draining and frustrating, in addition to being inconvenient, because they prevent him from spending time with his loved ones. Isaac also tracks flights for friends or family visiting him so he can plan pickups and adjust plans accordingly.
- **Goals:** What they want to accomplish
    - Understand the likelihood of a delay before purchasing a flight before departure
    - Know why a delay might or is happening
    - Provide alternative options for flights based on risk
    - Receive updates regarding loved ones' flights
- **Motivations:** Why they care / what success looks like

- o Wants to spend as much time as possible with loved ones
- o Prepare for all potential contingencies regarding travel
- o Aims for limited change in plans

- **Pain points:** What frustrates them today
  - o Flight tracking apps provide too limited context and additionally require payment
  - o Airlines provide notification regarding complications/delays/cancelations after the fact, without any updates or risk beforehand
  - o Weather apps require interpretation and don't easily provide insights to the likelihood of flight complications
  - o Challenging to compare different flights ahead of time and at the moment because there are so many different factors to consider amidst price and risk
- **Behaviors & habits:** Typical actions, routines, frequency
  - o Checks flight status and monitors weather conditions in the days before in the days before a flight
  - o Uses flight apps to track the flights of people visiting him
  - o Uses his phone to track flights, weather, and statuses, rarely on any other devices
- **Needs / requirements:** What the app must support
  - o Display the likelihood of complications before purchasing a flight and while a flight is purchased
  - o Provide real-time updates and notifications about spikes in delay/cancelation probability
  - o Suggest flights based on desired risk profile and cost
  - o Efficient cost
- **Quote:** A realistic sentence that captures their attitude
  - o "I hate the anxiety of not knowing if or when I'll be able to spend time with my loved ones."
- **Scenario / key task:** The main situation where they use your app
  - o Isaac will mainly use this app when making travel plans on his own to visit loved ones and as an aid when advising his loved ones when they are visiting him. Additionally, he'll use the app to track flights, weather, and upon traveling or receiving a visitor so he can make potential changes, be more aware of changes and challenges, and plan accordingly.

Optional (recommended): Review 5–10 app store reviews of a similar app and extract common complaints and wishes. You may use this to make your persona's pain points and goals more grounded in real user feedback.

3. **Usability Targets (20 pts)**

Define clear, measurable usability targets using both the ISO usability metrics and the Five Components of Usability. Your targets should be specific enough to test later (e.g., success criteria, time limits, error thresholds, and satisfaction ratings)

ISO Usability Metrics (10 pts)

- **Effectiveness:** What does "success" mean for your app? (Example: "User adds an assignment with a due date and sees it on the schedule.")
    - Search for any one-way, round-trip, or multi-city flight
    - View probability of a delay or cancelation for a potential or purchased flight
        - View the reasoning for the probability of delay/cancelation for a given flight
    - Provide potential alternative flights or airports to minimize cost and risk
        - For example, fly through DCA instead of IAD on 2/28 instead of 2/27.
    - Receive flight updates
    - At least 9 out of every 10 users should be able to either search for a current flight or a new flight, identify the risks associated with the flight, and easily identify potential alternatives.
- **Efficiency:** What does "fast enough" mean? (Example: "User adds an assignment and reminder in under 90 seconds.")
    - A new user can find a flight, identify the flight risks, and identify potential alternatives without lag within 3 minutes.
        - A returning user can do so within 1.5 minutes.
    - A user unfamiliar with technology is provided with a tutorial on how to complete the main tasks of the app that takes them 10 minutes.
- **Satisfaction:** How will you gauge it? (Example: "Post-task ease rating of 4+/5.")
    - First-time, returning, and those unfamiliar with technology will have the opportunity to rate their experience for a variety of questions on a scale of 1-5 (5 being the best).
        - How easy was it to use?
        - How understandable was it?
        - How efficient was the app?
        - How useful was the color scale for risk level?
        - How satisfactory was the design for you?

        - How likely are you to recommend this app to someone you know?
        - How would you rate your experience?
    - The average rating for all users (and within individual groups) must be at least 4 out of 5.

Five Components of Usability (10 pts)

- **Learnability:** How easy is it for new users to accomplish basic tasks?
    - 80% of first-time users can find or look up a flight, and identify the risks associated with that flight, without any help.
    - 95% of first-time users will understand the red/yellow/green color scale associated with risks
- **Efficiency:** Once learned, how quickly can users perform tasks?
    - Recurring users can look at prior flight search history and updates within 30 seconds
    - All users can find or look up a flight, and identify the risks associated with that flight.
- **Memorability:** How easily can returning users re-establish proficiency?
    - 90% of users can complete the app's main task (finding a flight and understanding risks) without consulting any help from any resources.
- **Errors:** How many errors occur, and how easily can users recover?
    - Potential errors include:
        - Selecting the incorrect flight
        - No flights available for a desired flight pattern
        - Misunderstanding flight risk
        - Not selecting all required fields for a search
        - No data available
        - Data is or models are inaccurate
    - Ideally, on average, a user would encounter an error less than 10% of the time, and in the case of certain errors, the app would provide guidance through why it's not working.
        - For example, if there are no flights available, the app will provide the user with potential flights from similar date ranges and close airports.
        - There will be notes for details for flight risks that users can look at in more detail.
        - If all fields are not selected, the app will prompt the user with what is missing.

        - If there is an error with the data or models, the app would immediately be under development to fix the issues.

- **Satisfaction:** How pleasant is the design to use?

    - Over 90% of answers for the survey question: "how satisfactory was the design for you?" are at least 4/5 or higher.

    - Users will describe the app as "easy", "helpful", "informative" or "simple".

4. **Core Design Principles (15 pts)**

Identify 3–5 design principles that you will intentionally apply to your application. For each principle, explain where/how you'll apply it in your app and why it matters for your persona. Your principles should directly connect to your persona's needs, habits, and pain points.

- **Progressive Disclosure**
    - Show essential information first and reveal detailed information only when the user asks for it.
    - Our app will display the delay "probability"/ risk color (red/yellow/green) immediately at the top of the flight results page. A collapsible "Why This Prediction?" The section will reveal detailed factors such as weather patterns, congestion, and historical performance.
    - Isaac wants quick clarity without cognitive overload. He often checks flights while he is busy with school. Progressive disclosure prevents overwhelming him with excessive detail while still allowing deeper insight when needed for decision-making.
- **Error Prevention**
    - Prevent users from making mistakes before they happen
    - Search button disabled until all required fields are completed. Clear airport autocomplete suggestions. Confirmation dialog before saving or replacing monitored flights. Clear labeling of time zones.
    - He often compares multiple airports and dates. Selecting the wrong airport or date could lead to poor decisions. Preventing errors reduces frustration and reinforces trust in the system.
- **Calm, Stress-Reducing Design**
    - Design that intentionally reduces stress through visual and interaction style.
    - Soft blue color palette inspired by clear skies. Minimal clutter and generous white space. Clear, conversational explanations of predictions.
    - Isaac's biggest pain point is travelling anxiety. The app should feel reassuring, not alarming. Emotional design supports the product's core value: stress-free travel planning.

5. **Heuristic Evaluation of a Similar Interface (10 pts)**

➔ Pick one comparable app or website (competitor or inspiration for your project)
➔ Conduct audit using Jakob Nielsen's 10 usability heuristics
➔ Report exactly 3 usability issues total

For each issue, include:

- Violated heuristic: Which heuristic (#1–#10) it breaks
- Why it matters: What friction, confusion, cognitive load, or errors it causes
- Severity rating (0–4): Use the scale below
- Suggested fix: One sentence 1

Severity rating scale:

- 0: Not a usability problem
- 1: Cosmetic only (fix if extra time)
- 2: Minor (low priority)
- 3: Major (high priority)
- 4: Catastrophe (must fix before release)

Comparable Interface: Flighty

- **Issue 1: High Information Density on Flight Detail Screen**
- **Violated Heuristic:**
  #8 – Aesthetic and Minimalist Design
- **Why it matters:**
  The flight detail screen presents multiple data points (live tracking, historical performance, aircraft details, airport congestion) simultaneously. For casual travelers, this increases cognitive load and makes it harder to quickly identify the most important insight: "Will this flight likely be delayed?"
- **Severity Rating:** 3 (Major)
- **Suggested Fix:**
  Prioritize delay probability and simplify secondary metrics behind expandable sections.

- **Issue 2: Lack of Clear Explanations for Predictions**
- **Violated Heuristic**
  #2 – Match Between System and the Real World
- **Why it matters:**
  Flighty often presents performance metrics and percentages without clearly explaining

what they mean in everyday language. Casual travelers may not understand statistical framing or aviation terminology, leading to confusion or misinterpretation.

- **Severity Rating:** 2 (Minor to Moderate)
- **Suggested Fix:**
  Provide plain-language explanations alongside technical metrics (

- **Issue 3: Subscription Paywall Blocks Core Functionality**
- **Violated Heuristic:**
  #3 – User Control and Freedom
- **Why it matters:**
  Key features are locked behind a subscription paywall. Users may explore the app but encounter blocked features mid-task, interrupting their flow and reducing perceived control over the experience.
- **Severity Rating:** 3 (Major)
- **Suggested Fix:**
  Provide limited free access to core prediction features or allow preview access before requiring subscription.

6. Usability Test Plan (15 pts)

You are planning your evaluation now; you will refine it in a later milestone. Your plan must include task-based usability testing (think-aloud) and one A/B test plan.

Part A: Task-Based Usability Testing (think-aloud)

- Write 3 realistic task scenarios based on your persona.
- Tasks must be goal-oriented (not step-by-step instructions) and neutral (do not hint what to click).
- Each task should reflect something your persona would genuinely do with your app.

For each task, specify the metrics you will measure:

- **Success/Fail:** Did the user complete the task? Define what counts as "success."
- **Time:** How long it took (include a target time aligned with your usability targets).
- **Errors:** Wrong actions, dead ends, incorrect inputs (count them).
- **Confusion:** Moments of uncertainty (pauses, rereading, backtracking, long stops).

Background 1: Isaac is travelling from Washington, D.C. to Minneapolis for the December holidays. Minnesota is notorious for bad weather during December, and he wants to know which flight has the lowest probability of being delayed or cancelled so he can make it home for Hanukkah.

Task 1: "Find the flight with the highest probability of being on time on December 16, 2026, from Washington D.C. to Minneapolis."
Success: Isaac selects a flight with the highest probability of being on time from any of the three major D.C. airports (DCA, IAD, or BWI) to Minneapolis/St. Paul (MSP).
Fail: Isaac doesn't find the flight with the highest on-time probability, or can't find the prediction
Time: The task should be completed in under 120 seconds.
Errors: Isaac could run into many errors, such as picking the wrong date, picking the wrong airport, or misinterpreting the flight's on-time probability
Confusion: Several things could cause Isaac to be confused, and therefore increase the amount of time it takes him to complete the task. These include not knowing which airport in D.C. to pick as the departure, and having trouble searching for the Minneapolis airport. Isaac could not find the filter button to sort flights by flight on-time probability and had to manually scroll through all options to find the flight with the highest probability.

Background 2: Isaac has booked a flight to New York City for an event. However, he needs to tell the host the probability of his flight being on time.
Task 2: "Check the on-time probability of Delta flight 442 from Ronald Reagan Washington National Airport to LaGuardia Airport on February 27, 2026."
Success: Isaac correctly calculates the probability that his specific flight will be on time.
Fail: Isaac is unable to find his specific flight, or can't find the on-time probability associated with his flight.
Time: The task should be completed in under 60 seconds.
Errors: Isaac could run into many errors, like entering the wrong date or airport for his flight.
Confusion: Several things could cause Isaac to be confused, and therefore increase the amount of time it takes for him to complete the task. These include having trouble searching for a specific flight or not knowing the airline code for Delta. Isaac could also have trouble finding the on-time probability for his specific flight.

Task 3: "Communicate why a flight from Washington D.C. to Miami has its assigned on-time probability."
Success: Isaac correctly explains why any flight from Washington, D.C., to Miami has an assigned probability.
Fail: Isaac is unable to or incorrectly explains why any flight from Washington, D.C., to Miami has an assigned probability.
Time: The task should be completed in under 90 seconds.
Errors: Isaac could run into many errors, including not selecting the right airports.
Confusion: Several things could cause Isaac to be confused, and therefore increase the amount of time it takes for him to complete the task. For instance, he could hesitate on which date to select. However, the most likely confusion is not knowing how to find the reason why a flight

has an assigned probability. Searching for this explanation could greatly extend the amount of time needed to complete this task.

Part B: A/B Test Plan

- Choose one specific design decision you are genuinely unsure about (e.g., navigation layout, button wording/placement, information density, default states, visual emphasis).
- Define Version A and Version B so the difference is unmistakable.
- State your hypothesis (which version you predict will perform better and why).
- Choose a metric to decide the winner (e.g., click-through rate, completion rate, time on task, error rate).
- Define success criteria (e.g., "Version B wins if it improves completion rate by ≥15%; otherwise choose the simpler/more consistent option.").

We are unsure how much information to provide AirAware users about why their selected flight might be delayed or cancelled. The goal of this "Prediction Explained" feature is to provide users with additional information to support their decision-making. However, we are unsure about whether or not this will be information overload for users. We have decided to run an A/B test to determine whether making this information immediately accessible will increase the time it takes for users to reach a decision outcome.

Version A will hide prediction information and require a user to select the "Explain this Prediction" button to view the reasoning behind the prediction. Version B will display this information upfront, without requiring the user to select a button.

We predict that Version B will perform better because many users will want to know why a potential flight might be delayed to further inform their decision. Requiring users to select a button (Version A) will only hamper this process and increase the time it takes to complete a task. The metric that we will use to judge these two versions is the completion time. Version B wins if the completion time is ≥20% faster than Version A.